



# Absolute Addressing





# Bakery Owner Dues to Suppliers

1. Repetition in contents
2. Occupies space
3. Effort to copy first row to next five rows.
4. When copying is to be performed to several rows, this model is not feasible.

| Prices     |         |         |         |  |
|------------|---------|---------|---------|--|
|            | Sugar   | Butter  | Flour   |  |
| Supplier 1 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 2 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 3 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 4 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 5 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
| Supplier 6 | \$ 0.40 | \$ 1.20 | \$ 0.12 |  |
|            |         |         |         |  |
| Quantity   |         |         |         |  |
|            | Sugar   | Butter  | Flour   |  |
| Supplier 1 | 364     | 391     | 220     |  |
| Supplier 2 | 387     | 245     | 314     |  |
| Supplier 3 | 290     | 211     | 200     |  |
| Supplier 4 | 340     | 265     | 330     |  |
| Supplier 5 | 261     | 345     | 246     |  |
| Supplier 6 | 365     | 232     | 390     |  |

# Bakery Owner Dues to Suppliers – Alternate model



|            | Price  |        |        |
|------------|--------|--------|--------|
|            | \$0.40 | \$1.20 | \$0.12 |
| Quantity   | Sugar  | Butter | Flour  |
| Supplier 1 | 364    | 391    | 220    |
| Supplier 2 | 387    | 245    | 314    |
| Supplier 3 | 290    | 211    | 200    |
| Supplier 4 | 340    | 265    | 330    |
| Supplier 5 | 261    | 345    | 246    |
| Supplier 6 | 365    | 232    | 390    |

1. In this model, repetition issues has been addressed
2. Space issue is addressed
3. No need to duplicate the prices data
4. As far as input data model representation is concerned, this is a very good model.
5. This model causes problems in computation and particularly when copying the formula

# Bakery Owner Dues to Suppliers – Alternate model



|   | A          | B       | C      | D     |
|---|------------|---------|--------|-------|
| 1 |            | Sugar   | Butter | Flour |
| 2 |            | 0.40    | 1.20   | 0.12  |
| 3 | Supplier 1 | 364     | 391    | 220   |
| 4 | Supplier 2 | 387     | 245    | 314   |
| 5 |            |         |        |       |
| 6 | Formula    | B2 * B3 |        |       |
| 7 |            |         |        |       |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B              | C      | D     |
|---|----------------|----------------|--------|-------|
| 1 |                | Sugar          | Butter | Flour |
| 2 |                | 0.40           | 1.20   | 0.12  |
| 3 | Supplier 1     | <b>364</b>     | 391    | 220   |
| 4 | Supplier 2     | <b>387</b>     | 245    | 314   |
| 5 |                |                |        |       |
| 6 | Formula        | B2 * B3        |        |       |
| 7 | Copied formula | <b>B3</b> * B4 |        |       |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C           | D     |
|---|----------------|-----------|-------------|-------|
| 1 |                | Sugar     | Butter      | Flour |
| 2 |                | 0.40      | <b>1.20</b> | 0.12  |
| 3 | Supplier 1     | 364       | <b>391</b>  | 220   |
| 4 | Supplier 2     | 387       | 245         | 314   |
| 5 |                |           |             |       |
| 6 | Formula        | $B2 * B3$ | $C2 * C3$   |       |
| 7 | Copied formula | $B3 * B4$ |             |       |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C              | D     |
|---|----------------|-----------|----------------|-------|
| 1 |                | Sugar     | Butter         | Flour |
| 2 |                | 0.40      | 1.20           | 0.12  |
| 3 | Supplier 1     | 364       | <b>391</b>     | 220   |
| 4 | Supplier 2     | 387       | <b>245</b>     | 314   |
| 5 |                |           |                |       |
| 6 | Formula        | $B2 * B3$ | $C2 * C3$      |       |
| 7 | Copied formula | $B3 * B4$ | <b>C3</b> * C4 |       |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C         | D           |
|---|----------------|-----------|-----------|-------------|
| 1 |                | Sugar     | Butter    | Flour       |
| 2 |                | 0.40      | 1.20      | <b>0.12</b> |
| 3 | Supplier 1     | 364       | 391       | <b>220</b>  |
| 4 | Supplier 2     | 387       | 245       | 314         |
| 5 |                |           |           |             |
| 6 | Formula        | $B2 * B3$ | $C2 * C3$ | $D2 * D3$   |
| 7 | Copied formula | $B3 * B4$ | $C3 * C4$ |             |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C         | D         |
|---|----------------|-----------|-----------|-----------|
| 1 |                | Sugar     | Butter    | Flour     |
| 2 |                | 0.40      | 1.20      | 0.12      |
| 3 | Supplier 1     | 364       | 391       | 220       |
| 4 | Supplier 2     | 387       | 245       | 314       |
| 5 |                |           |           |           |
| 6 | Formula        | $B2 * B3$ | $C2 * C3$ | $D2 * D3$ |
| 7 | Copied formula | $B3 * B4$ | $C3 * C4$ | $D3 * D4$ |

**Requirement is:**  
**While copying**  
**formula, price per**  
**unit quantity**  
**cell SHOULD NO**  
**T change.**



## **Absolute Addressing – Varying Column, Fixed Row**



# Varying Column, Fixed Row and Effect of Copy



- In order for copy formula not to change row number, place **DOLLAR** sign in front of row number alone.

|   | A              | B                | C                | D                |
|---|----------------|------------------|------------------|------------------|
| 1 |                | Sugar            | Butter           | Flour            |
| 2 |                | 0.40             | 1.20             | 0.12             |
| 3 | Supplier 1     | 364              | 391              | 220              |
| 4 | Supplier 2     | 387              | 245              | 314              |
| 5 |                |                  |                  |                  |
| 6 | Formula        | <b>B\$2</b> * B3 | <b>C\$2</b> * C3 | <b>D\$2</b> * D3 |
| 7 | Copied formula | <b>B\$2</b> * B4 | <b>C\$2</b> * C4 | <b>D\$2</b> * D4 |

# Varying Column, Fixed Row and Effect of Copy



- In order for copy formula not to change row number, place DOLLAR sign in front of row alone.
- Example: Cell I8 contains the formula D\$5
- When the formula in cell I8 is copied to different rows of column I, say cells I9 and I10
  - As DOLLAR sign is in front of row number of the formula, the row number is fixed to 5
  - As the formula is copied to same column D, column name is D.
  - Therefore, the copied formula is D\$5 in both I9 and I10.



## **Absolute Addressing – Fixed Column, Varying Row**



# Fixed Column , Varying Row and Effect of Copy



- In order for copy formula not to change column name, place DOLLAR sign in front column name.
- Example: Cell I8 contains the formula \$D5
- When formula in cell I8 is copied to cells I9 and I10,
  - As there is DOLLAR sign in column name, column name is NOT varied. It is D
  - As there is NO DOLLAR sign in row number is incremented to 6 and 7
  - There the formula copied in cells I9 and I10 are \$D6, \$D7



## **Absolute Addressing – Fixed Column, Fixed Row**



# Fixed Column, Fixed Row and Effect of Copy



- Cell is referred as (Column Name, Row Number).
- In order for copy formula not to change column name, place DOLLAR sign in front column name.
- In order for copy formula not to change row number, place DOLLAR sign in front row number.
- Example: Cell I8 contains the formula `$D$5`
- When formula in cell I8 is copied to cell I6 and I7,
  - As there is DOLLAR sign appears in column name and row number, no change is made to formula and `$D$5` is copied in I6 and I7 cells

# Bakery Owner Dues to Suppliers – Modified Formula



|   | A          | B         | C      | D     |
|---|------------|-----------|--------|-------|
| 1 |            | Sugar     | Butter | Flour |
| 2 |            | 0.40      | 1.20   | 0.12  |
| 3 | Supplier 1 | 364       | 391    | 220   |
| 4 | Supplier 2 | 387       | 245    | 314   |
| 5 |            |           |        |       |
| 6 | Formula    | B\$2 * B3 |        |       |
| 7 |            |           |        |       |

# Bakery Owner Dues to Suppliers – Modified Formula



|   | A              | B                | C      | D     |
|---|----------------|------------------|--------|-------|
| 1 |                | Sugar            | Butter | Flour |
| 2 |                | 0.40             | 1.20   | 0.12  |
| 3 | Supplier 1     | <b>364</b>       | 391    | 220   |
| 4 | Supplier 2     | <b>387</b>       | 245    | 314   |
| 5 |                |                  |        |       |
| 6 | Formula        | B\$2 * B3        |        |       |
| 7 | Copied formula | <b>B\$2</b> * B4 |        |       |

# Bakery Owner Dues to Suppliers – Modified Formula



|   | A              | B       | C         | D     |
|---|----------------|---------|-----------|-------|
| 1 |                | Sugar   | Butter    | Flour |
| 2 |                | 0.40    | 1.20      | 0.12  |
| 3 | Supplier 1     | 364     | 391       | 220   |
| 4 | Supplier 2     | 387     | 245       | 314   |
| 5 |                |         |           |       |
| 6 | Formula        | B2 * B3 | C\$2 * C3 |       |
| 7 | Copied formula | B3 * B4 |           |       |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C                | D     |
|---|----------------|-----------|------------------|-------|
| 1 |                | Sugar     | Butter           | Flour |
| 2 |                | 0.40      | 1.20             | 0.12  |
| 3 | Supplier 1     | 364       | <b>391</b>       | 220   |
| 4 | Supplier 2     | 387       | <b>245</b>       | 314   |
| 5 |                |           |                  |       |
| 6 | Formula        | B\$2 * B3 | C\$2 * C3        |       |
| 7 | Copied formula | B\$2 * B4 | <b>C\$2</b> * C4 |       |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C           | D                |
|---|----------------|-----------|-------------|------------------|
| 1 |                | Sugar     | Butter      | Flour            |
| 2 |                | 0.40      | <b>1.20</b> | <b>0.12</b>      |
| 3 | Supplier 1     | 364       | <b>391</b>  | <b>220</b>       |
| 4 | Supplier 2     | 387       | 245         | 314              |
| 5 |                |           |             |                  |
| 6 | Formula        | B\$2 * B3 | C\$2 * C3   | <b>D\$2 * D3</b> |
| 7 | Copied formula | B\$3 * B4 | C\$2 * C4   |                  |

# Bakery Owner Dues to Suppliers – Alternate model



|   | A              | B         | C         | D         |
|---|----------------|-----------|-----------|-----------|
| 1 |                | Sugar     | Butter    | Flour     |
| 2 |                | 0.40      | 1.20      | 0.12      |
| 3 | Supplier 1     | 364       | 391       | 220       |
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| 5 |                |           |           |           |
| 6 | Formula        | B\$2 * B3 | C\$2 * C3 | D\$2 * D3 |
| 7 | Copied formula | B\$3 * B4 | C\$2 * C4 | D\$2 * D4 |

**Requirement is:**  
While copying  
formula, the row  
number  
**SHOULD NOT**  
change.





# **Analysis - Demand Curve Estimation**



# Coffee Demand Curve Estimation



- To set up a new coffee shop, the owner is planning to price the coffee per cup.
- The owner knows demand for his coffee in terms of number of coffee cups expected to be sold every day
- He also knows the cost of producing coffee per cup (production price).
- He want to understand at what price he must sell coffee per cup that gives him maximum profit.
- That is what is the selling price that gives him maximum profit.



# Coffee Demand Curve Estimation

- **Given inputs**
  - Demand in terms of number of cups =  $(100 - 15 * \text{selling price})$
  - Profit =  $(\text{selling price} - \text{production price}) * \text{Demand}$
- **Estimate the profit by**
  - Varying the production price from \$0.5 to \$2.0 and
  - Varying the selling price from \$2.0 to \$5.0
  - Compute the profit



# Coffee Demand Curve Estimation - Example

$$\text{Demand} = (100 - 15 * \text{selling price})$$

| Production price | Selling price | Demand              | Profit                 |
|------------------|---------------|---------------------|------------------------|
| 0.5              | 2.0           | $100 - 15 * 2 = 70$ | $(2 - 0.5) * 70 = 105$ |
| 1.0              | 2.0           | $100 - 15 * 2 = 70$ | $(2 - 1.0) * 70 = 70$  |
| 1.5              | 2.0           | $100 - 15 * 2 = 70$ | $(2 - 1.5) * 70 = 35$  |
| 2.0              | 2.0           | $100 - 15 * 2 = 70$ | $(2 - 2) * 70 = 0$     |



# Coffee Demand Curve Estimation - Example

$$\text{Demand} = (100 - 15 * \text{selling price})$$

| Production Price | Selling Price | Demand                  | Profit                       |
|------------------|---------------|-------------------------|------------------------------|
| 0.5              | 2.5           | $100 - 15 * 2.5 = 62.5$ | $(2.5 - 0.5) * 62.5 = 125$   |
| 1.0              | 2.5           | $100 - 15 * 2.5 = 62.5$ | $(2.5 - 1.0) * 62.5 = 93.75$ |
| 1.5              | 2.5           | $100 - 15 * 2.5 = 62.5$ | $(2.5 - 1.5) * 62.5 = 62.5$  |
| 2.0              | 2.5           | $100 - 15 * 2.5 = 62.5$ | $(2.5 - 2) * 62.5 = 31.25$   |





## **Naming Group of Cells – Range Name**



# Range Names



- Cells are referred using (Column Name Row Number) notation
- A1, B7, D12 etc
- Group of cells are referred using colon notation
  - Rows 1 to 100: A1:A100
  - Columns B to G: B7:G7
  - Rows & Columns: A1:C3 (A1, A2, A3, B1, B2, B3, C1, C2 and C3)
- It is easier if cells are given name instead of A1, B7, D12 or name to group of cells such as marks instead of A1:A100
- Name to cells are given in the interface Name box



**Name individual cells**





## **Name range of cells**





## **Example – 01 – State-wise Sales**





## **Example – 02 – Historical Investment**







**Thank You!**

